Chipsee

Industrial PC

CS-JETORIN-BOX

PN: CS-JETORIN-BOX

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Contents

CS-JETORIN-BOX

Front View

Rear View

Side View 1

Side View 2

Product Overview

The PPC-JETORIN series CS-JETORIN-BOX is a rugged, high-quality industrial box PC.

Key Applications

- Human Machine Interface HMI
- Industrial Automation
- Process Control
- Smart Grid Management
- CNC Manufacturing
- Environmental Monitoring
- Machine Vision Inspection
- ATM, Kiosk, Infotainment...

The CS-JETORIN-BOX Industrial Box PC supports four Jetson module variants: **NVIDIA Jetson Orin Nano 4GB/8GB and NVIDIA Jetson Orin NX 8GB/16GB**. Equipped with a broad range of connectivity options, it satisfies demanding application requirements in harsh industrial and outdoor environments.

A specially designed magnesium-aluminum alloy housing with fins for increased heat dissipation serves as a passive cooler, eliminating the need for built-in fans. The fan-less design reduces noise, as well as the maintenance costs and efforts, increasing reliability at the same time.

> **Caution**
>
> Be careful when handling the product while it is operating: the back panel might become hot under heavy CPU load.

Ordering Options

Most of the Chipsee products can be customized during the ordering process. The product will be shipped with the pre-installed factory defaults if no extra requirements are specified. The table in the Hardware Features section provides information about the default options bundled with the product.

> **Note**
>
> You can order CS-JETORIN-BOX from the official Chipsee Store or from your nearest distributor.

Operating System

By default, CS-JETORIN-BOX comes with Linux(Ubuntu 24.04) pre-installed.

Optional Features

Jetson Orin Module

The Jetson Orin Module appears in different variants (**NVIDIA Jetson Orin Nano 4GB/8GB and NVIDIA Jetson Orin NX 8GB/16GB**).

The CS-JETORIN-BOX Industrial Box PC does not include the Jetson Orin Module by default.

If you would like to purchase it with a Jetson Orin Module, you can select it at the Chipsee store during the ordering process.

Item	Jetson Orin Nano 4GB	Jetson Orin Nano 8GB	Jetson Orin NX 8GB	Jetson Orin NX 16GB
AI Performance	34 TOPS	67 TOPS	117 TOPS	157 TOPS
GPU	512 core NVIDIA Ampere GPU with 16 Tensor Cores	1024 core NVIDIA Ampere GPU with 32 Tensor Cores	1024 core NVIDIA Ampere GPU with 32 Tensor Cores	1024 core NVIDIA Ampere GPU with 32 Tensor Cores
GPU Max Frequency	1020 MHz	1020 MHz	1173 MHz	1173 MHz
CPU	6 core Arm® Cortex® A78AE v8.2 64 bit CPU 1.5 MB L2 + 4 MB L3	6 core Arm® Cortex® A78AE v8.2 64 bit CPU 1.5 MB L2 + 4 MB L3	6 core Arm® Cortex® A78AE v8.2 64 bit CPU 1.5 MB L2 + 4 MB L3	8 core Arm® Cortex® A78AE v8.2 64 bit CPU 2 MB L2 + 4 MB L3
CPU Max Frequency	1.7 GHz	1.7 GHz	2 GHz	2 GHz
DL Accelerator	-	-	1x NVDLA v2	2x NVDLA v2
DLA Max Frequency	-	-	1230 MHz	1230 MHz
Memory	4GB 64 bit LPDDR5, 51 GB/s	8GB 128 bit LPDDR5, 102 GB/s	8GB 128 bit LPDDR5, 102 GB/s	16GB 128 bit LPDDR5, 102.4 GB/s
Storage	(Supports external NVMe)	(Supports external NVMe)	(Supports external NVMe)	(Supports external NVMe)
Video Encode	1080p30 supported by 1-2 CPU cores	1080p30 supported by 1-2 CPU cores	1x 4K60(H.265), 3x 4K30 (H.265), 6x 1080p60(H.265), 12x 1080p30(H.265)	1x 4K60(H.265), 3x 4K30 (H.265), 6x 1080p60(H.265), 12x 1080p30(H.265)
Video Decode	1x 4K60(H.265), 2x 4K30(H.265), 5x 1080p60(H.265),	1x 4K60(H.265), 2x 4K30(H.265), 5x 1080p60(H.265),	1x 8K30(H.265), 2x 4K60(H.265), 4x 4K30(H.265), 9x	1x 8K30(H.265), 2x 4K60(H.265), 4x 4K30(H.265), 9x

Item	Jetson Orin Nano 4GB	Jetson Orin Nano 8GB	Jetson Orin NX 8GB	Jetson Orin NX 16GB
	11x 1080p30(H. 265)	11x 1080p30(H. 265)	1080p60(H.265), 18x 1080p30(H. 265)	1080p60(H.265), 18x 1080p30(H. 265)

Table 510 Jetson Orin Nano / NX Series Specifications

4G/5G/WiFi/BT Module

The CS-JETORIN-BOX Industrial Box PC does not include WiFi/BT and/or 4G/5G modules by default. These modules are optional and can be selected at the Chipsee store during the ordering process.

The 4G/5G module only supports USB2.0 speed.

> **Warning**
>
> Installation, repair, and maintenance tasks should be performed by trained personnel only.
> Chipsee does not bear any responsibility for damage caused by inadequate handling of the product.

Hardware Features

The CS-JETORIN-BOX Industrial Box PC offers a broad range of performance and connectivity options for scalable integration, providing expandability to fit future requirements. Some of the key features are listed in the table below.

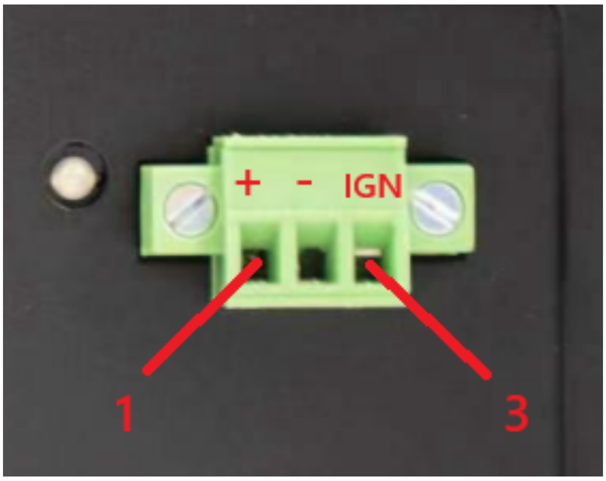
CS-JETORIN-BOX	
Display	N/A
Storage	Default mSATA 128GB SSD
Touch	N/A
Audio	N/A
USB	6 x USB 3.0 Type A HOST and 1 x Type C port
LAN	2 x RJ45, 1Gbps
UART	Default 2 x RS232(include one debug port). 1 x RS232 can be configured to 1 x RS485(optional)
GPIO	N/A
4G/5G	Optional, module available from the Chipsee store
WiFi/BT	Optional, module available from the Chipsee store
I2C	2 x I2C port
HDMI	1 x HDMI 2.0 out
Power IN	from 15V to 48V DC
Power Consumption	30W Max with Nano 4GB/8GB, 45W Max with NX 4GB/8GB
OS	Ubuntu 24.04(default)
Operating Temp.	From 0°C to +60°C
Storage Temp.	From +18°C to +30°C
Humidity	From 5% to 95% relative humidity (no condensation)
Dimensions	158 X 118 X 68mm
EMC	CE/FCC/ROHS/ISO-9001
Mounting	VESA 100mm, Desktop
Weight	1200g

Table 511 Key Features

Power Input

The CS-JETORIN-BOX Industrial Box PC can be powered by a wide range of input voltages: from 15V to 48V DC.

It is a **3 Pin, 3.81mm screw terminal** connector. As shown in the figure below.



Power Input

Note that the "**+**" sign represents the positive power input. The "**-**" terminal is shorted to the ground. The product has a "ignition signal" **optional** feature. By default the ignition signal is not installed. If you need this feature you can contact us when placing an order. In this setup, Pin 3 is the ignition signal pin. To use this feature, apply a 12V or 24V DC input (relative to -) to Pin 3. If Pin 3 detects a low input voltage, the product will be shutdown. If Pin 3 detects a high input voltage, the product will be boot and running.

Power Input Definition		
Pin Number	**Definition**	**Description**
Pin 1	Positive Input	DC Power **Positive Terminal**
Pin 2	Negative Input	DC Power **Negative Terminal**
Pin 3	IGN	**Ignition Signal**

Table 512 Power Connector

Note

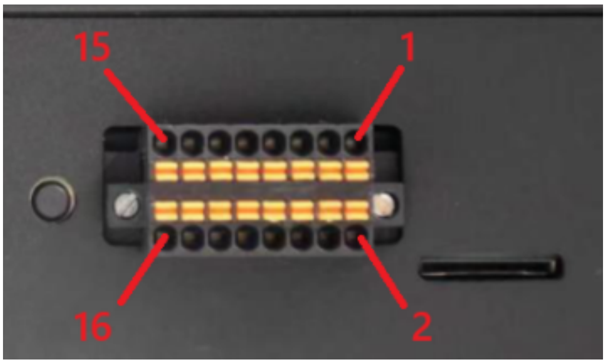
The ignition signal "**IGN**" is an optional feature, it is not mounted default.

Connectivity

There are many connectivity options available on the CS-JETORIN-BOX industrial PC. It has 6 x USB Type A connectors configured as HOSTS, 1 x Type C connector, 2 x RJ45 connectors supporting Gigabit Ethernet (GbE), and up to 2 x RS232 connectors(include 1 x Debug port), one of RS232 can be configured to 1 x RS485, 1 x Fakra-Mini connector.

RS232/485/I2C/CAN Connectors

The CS-JETORIN-BOX Industrial Box PC has 2 x 8-pin 2.54mm connector with pluggable terminal block.



RS232,RS485,I2C,CAN

Pin	Signal Name	Pin	Signal Name
1	I2C7_SCL_3V3	2	I2C7_SDA_3V3
3	I2C1_SCL_3V3	4	I2C1_SDA_3V3
5	CAN1_L (-)	6	CAN1_H (+)
7	CAN0_L (-)	8	CAN0_H (+)
9	RS485_1+ (A)	10	RS485_1- (B)
11	RS232_1_TXD	12	RS232_1_RXD
13	RS232_2_TXD	14	RS232_2_RXD
15	GND	16	GND

Attention

- Pins 9/10 (RS485_1) and pins 11/12 (RS232_1) share one hardware multiplexed channel. **Only one interface can be selected for use at a time, cannot work simultaneously.**
- The RS232_1 is enabled and RS485_1 is disabled default.
- The 120Ω match resistor for **CAN** bus is **NOT mounted** by default.
- The 120Ω match resistor for **RS485** bus is **NOT mounted** by default.

Fakra-Mini Connectors

The CS-JETORIN-BOX Industrial Box PC supports Fakra-Mini automotive connector as shown in the figure below.

Fakra-Mini automotive Connectors

Fakra-Mini automotive connector, driven by MAX96724 GMSL-2 serializer chip. It outputs four independent GMSL-2 single-ended serial signals, used to connect remote automotive cameras via coaxial cables with PoC power-over-coax.

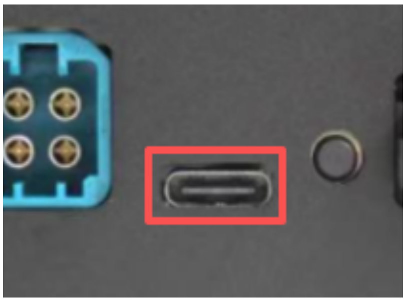
Pin	Net Name	Description
1	A0P	GMSL-2 Serial Lane A positive output
2	B0P	GMSL-2 Serial Lane B positive output
3	C0P	GMSL-2 Serial Lane C positive output
4	D0P	GMSL-2 Serial Lane D positive output

Table 513 FAKRA-MINI Connector Pinout

USB HOST Connectors

CS-JETORIN-BOX is equipped with 7 USB connectors. One is a USB2.0 Type C connector supporting device, and USB Recovery. In addition, there are three, dual stacked USB 3.0 Type A connectors. Each connector supports host mode only.

6 x USB3.0 HOST Connectors



1 x TypeC Connectors

HDMI Connector

The CS-JETORIN-BOX Industrial Box PC is equipped with 1 x **HDMI** connector. The HDMI connector allows connecting an additional (external) monitor. HDMI output resolution can be configured by the software.

HDMI Connector

LAN Connectors

2 x **LAN (RJ45) connectors** provide Ethernet connectivity over standardized Ethernet cables. The integrated two-port Ethernet interface supports 10/100/1000BASE-T/TX specifications with automatic speed negotiation feature. Power over Ethernet (PoE) is **not** supported.

2 x RJ45 GbE LAN Connectors

> **Note**
>
> Use CAT5 or better cables to achieve full data throughput over maximum distance defined by the 1000BASE-T standard (100m).

WiFi & BlueTooth Module

The CS-JETORIN-BOX Industrial Box PC is equipped with the popular **SKO.WBM80X2P.2 WiFi/BT module** which supports BT/BLE 2.1/3.0/4.x/5.3/5.4, as well as 802.11ax/ac/a/b/g/n 600Mbps 2.4/5.8 GHz Wireless LAN (WLAN).

SKO.WBM80X2P.2 WiFi/BT Module

The CS-JETORIN-BOX includes an SMA connector for an external WiFi/BT antenna, as illustrated in the figure below.

WiFi+BT Antenna SMA

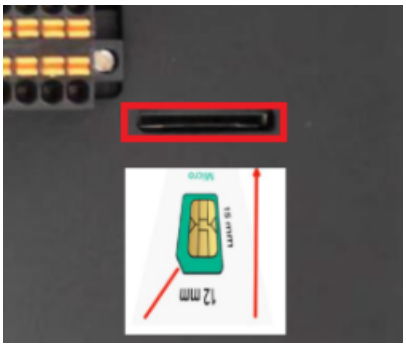
4G/LTE Module

The CS-JETORIN-BOX Industrial Box PC is equipped with a M.2 connector that can connect an optional 4G/LTE module. The customer will also need a SIM Card Holder and a 4G/LTE antenna connector to ensure 4G/LTE works.

SIM card does **NOT** support hot plug. **Power off** before inserting or removing SIM card.

M.2 Connector & 4G Module

4G/LTE Antenna



SIM Card Direction

Attention

> The product does not come shipped with the 4G/LTE module by default. The customer can choose the 4G/LTE module option when placing an order, we will install all the necessary components.

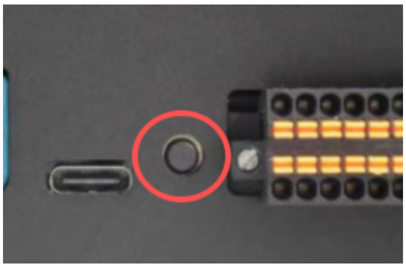
Recovery Button

The CS-JETORIN-BOX Industrial Box PC has one button on the board marked as **Recovery**, as shown in the figure below.

When the button is pressed before powering up, the CS-JETORIN-BOX will enter **Recovery** mode. In this mode you can use a USB Type-C cable to upgrade its operating system. You can use this feature to flash another OS to the internal NVME SSD.

When the button is not pressed before and during power up, the CS-JETORIN-BOX will boot normally.

There is no need to press the button during regular operation. However, if you need to flash the OS in **Recovery** mode, the button will be used. Please refer to the software documents for more information.



Recovery Button

Mounting Procedure

The CS-JETORIN-BOX Industrial Box PC supports VESA 100 x 100 mounting pattern with 4 x M4 screws, enabling simplified installation onto any standard VESA mounting rack.

The CS-JETORIN-BOX Industrial Box PC supports panel mount, wall mount and desktop.

You can find detailed information about mounting in the Mount IPC Guide.

Mechanical Specifications

The outer mechanical dimensions of The CS-JETORIN-BOX Industrial Box PC are 158 X 118 X 68mm (W x L x H). Please refer to the technical drawing in the figure below for details related to the specific product measurements.

Technical Drawing and Cutout Dimensions

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